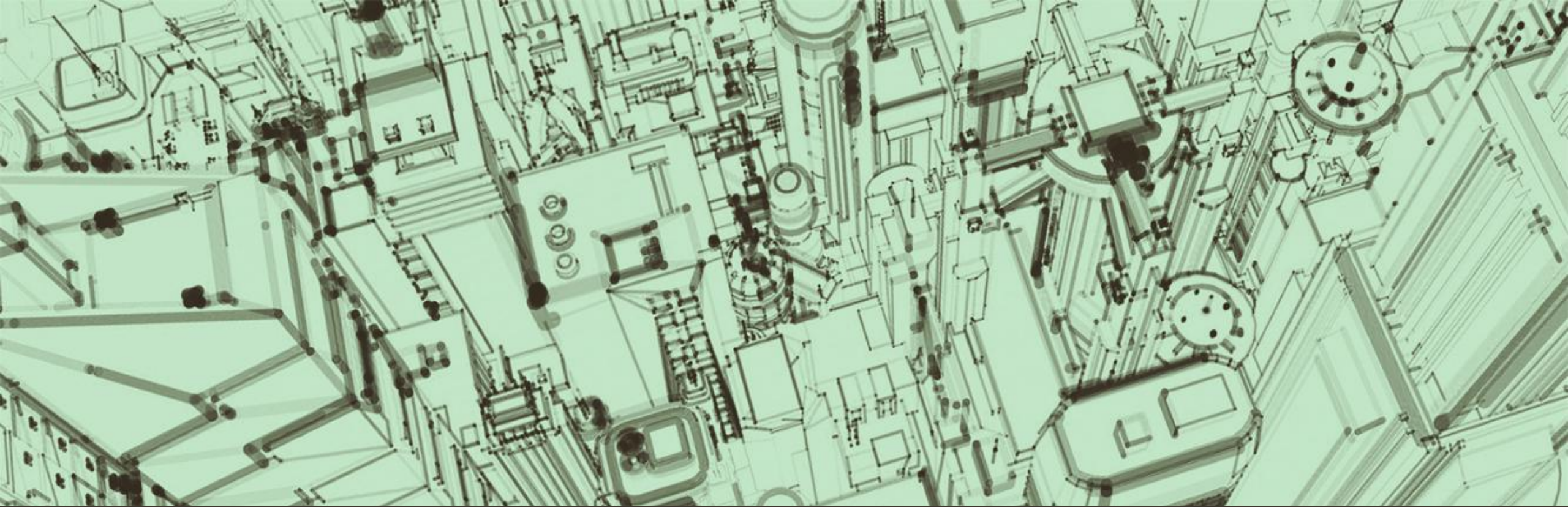




# Software Engineering



- Load Testing using Jmeter
- Cost Estimation Functional Point

# Step 1 – Install Java



## Step 2 – Download JMeter

### Apache JMeter 5.1 (Requires Java 8+)

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#### Binaries

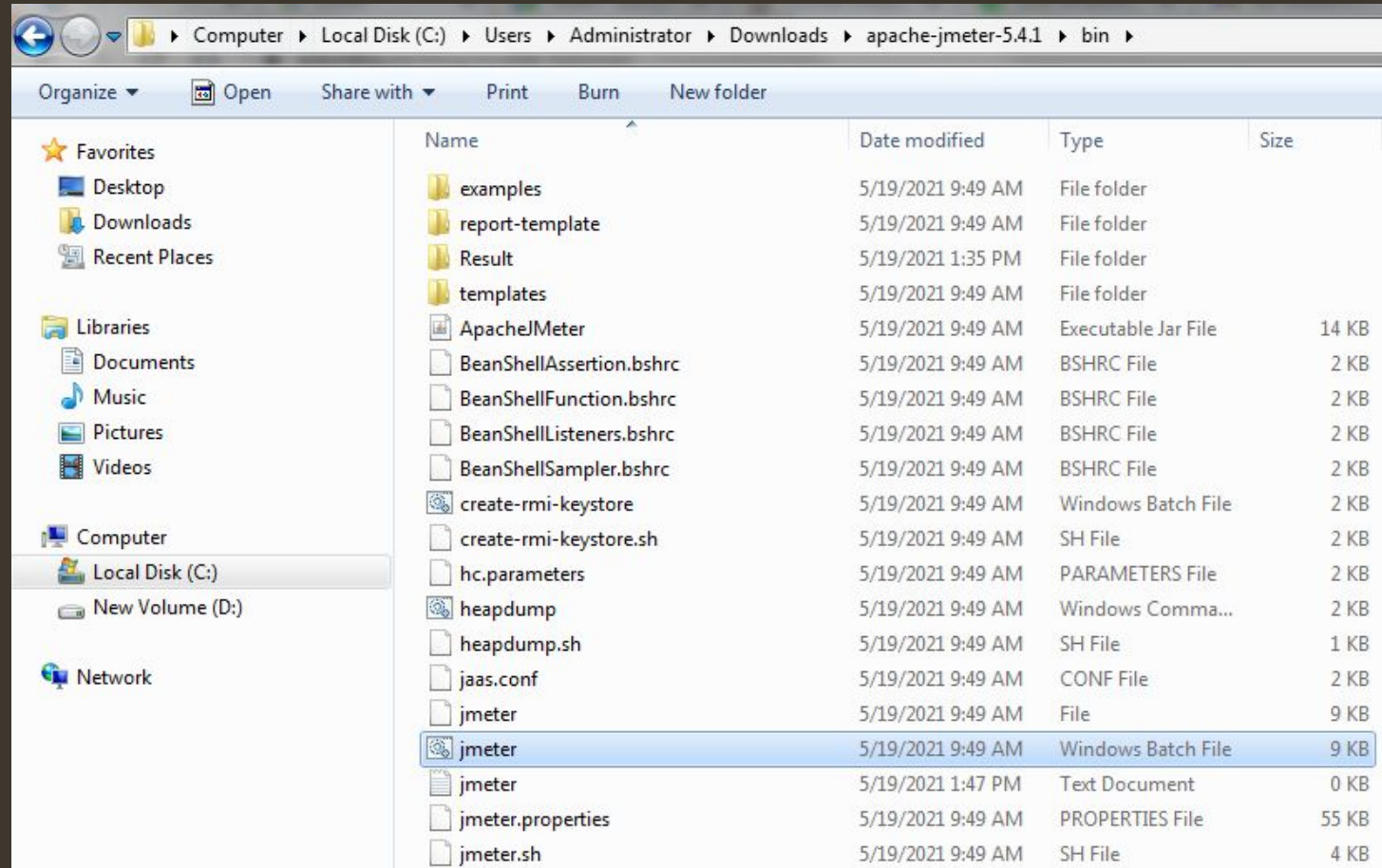
---

|                                                                                  |
|----------------------------------------------------------------------------------|
| <a href="#">apache-jmeter-5.1.tgz</a> <a href="#">sha512</a> <a href="#">pgp</a> |
| <a href="#">apache-jmeter-5.1.zip</a> <a href="#">sha512</a> <a href="#">pgp</a> |

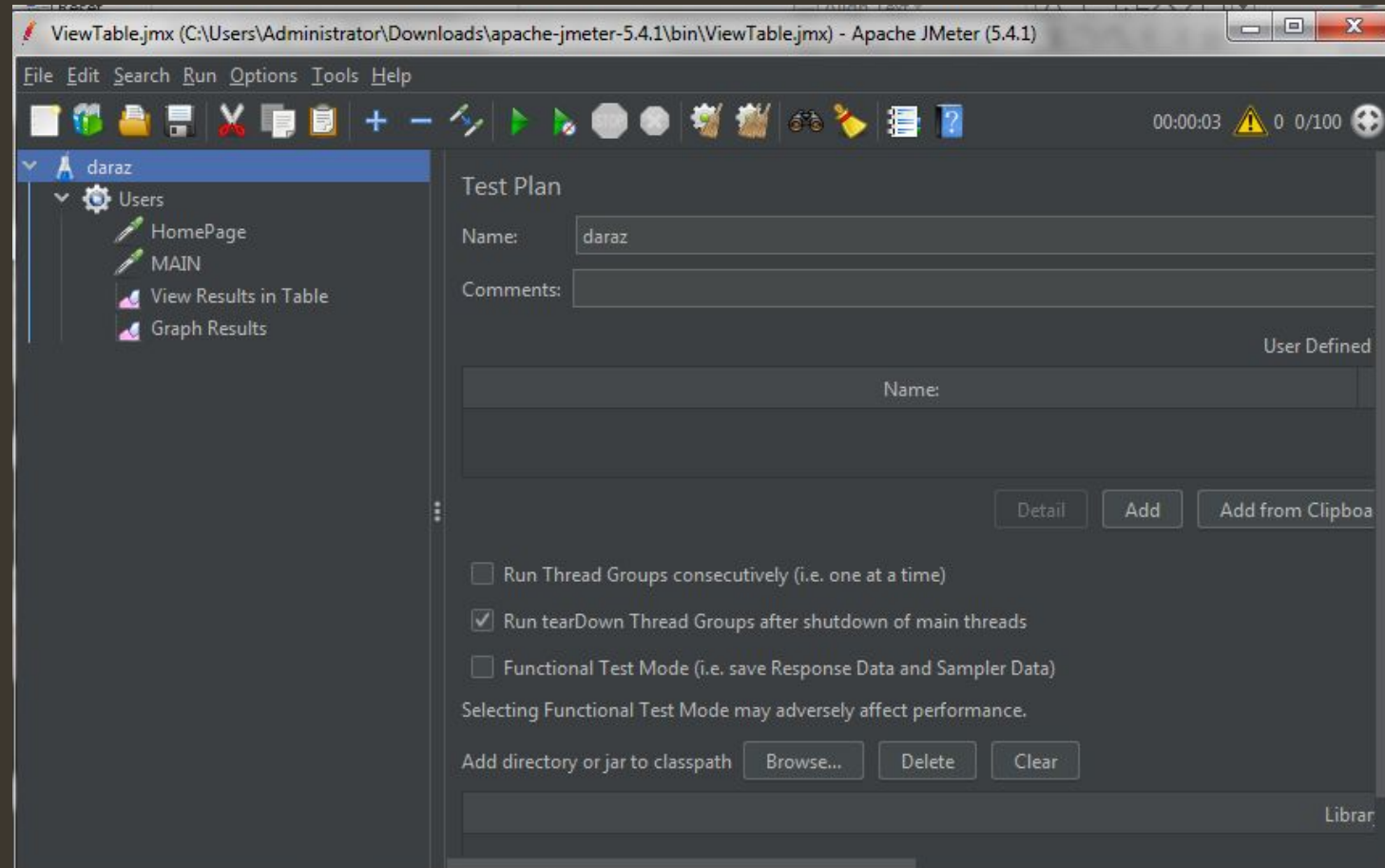
[https://jmeter.apache.org/download\\_jmeter.cgi](https://jmeter.apache.org/download_jmeter.cgi)



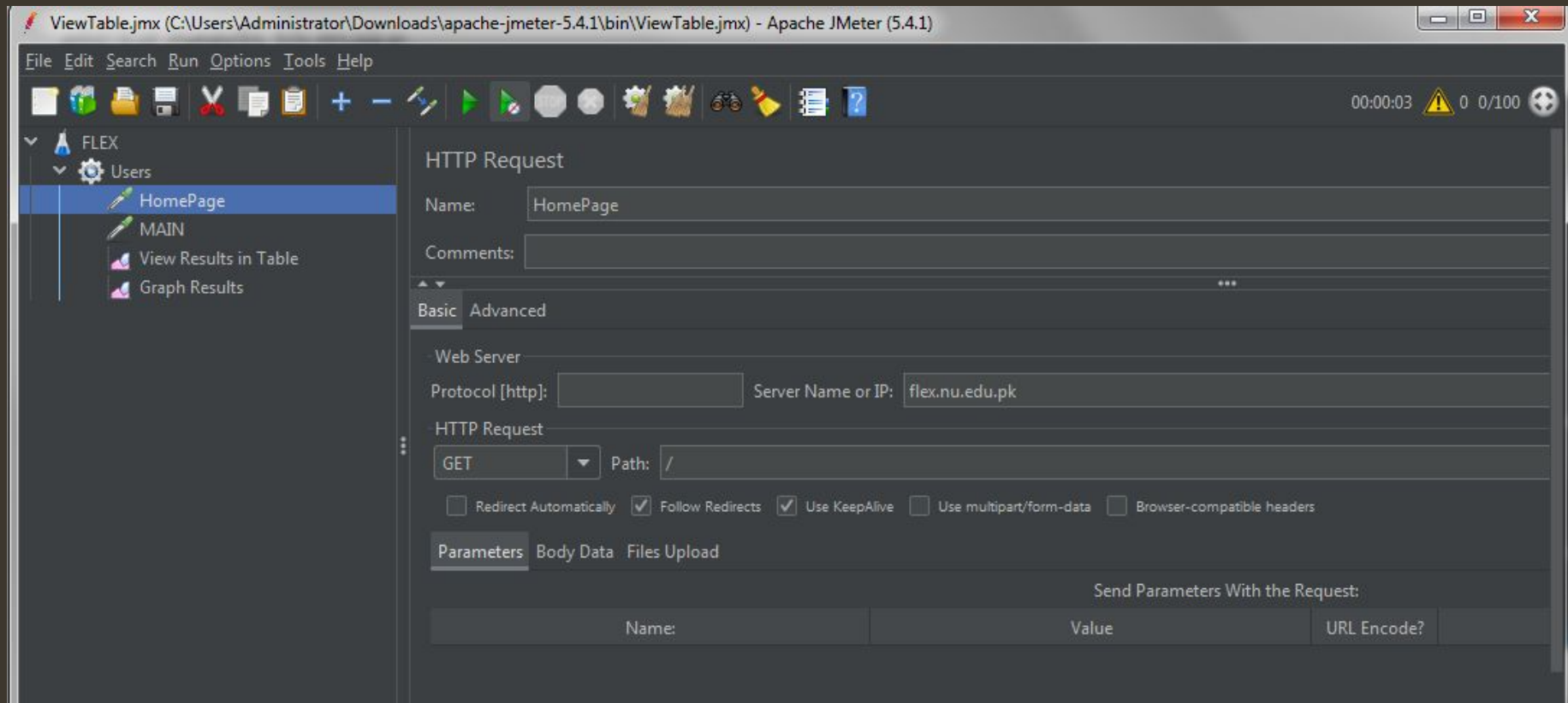
# Step 3 – Install JMeter



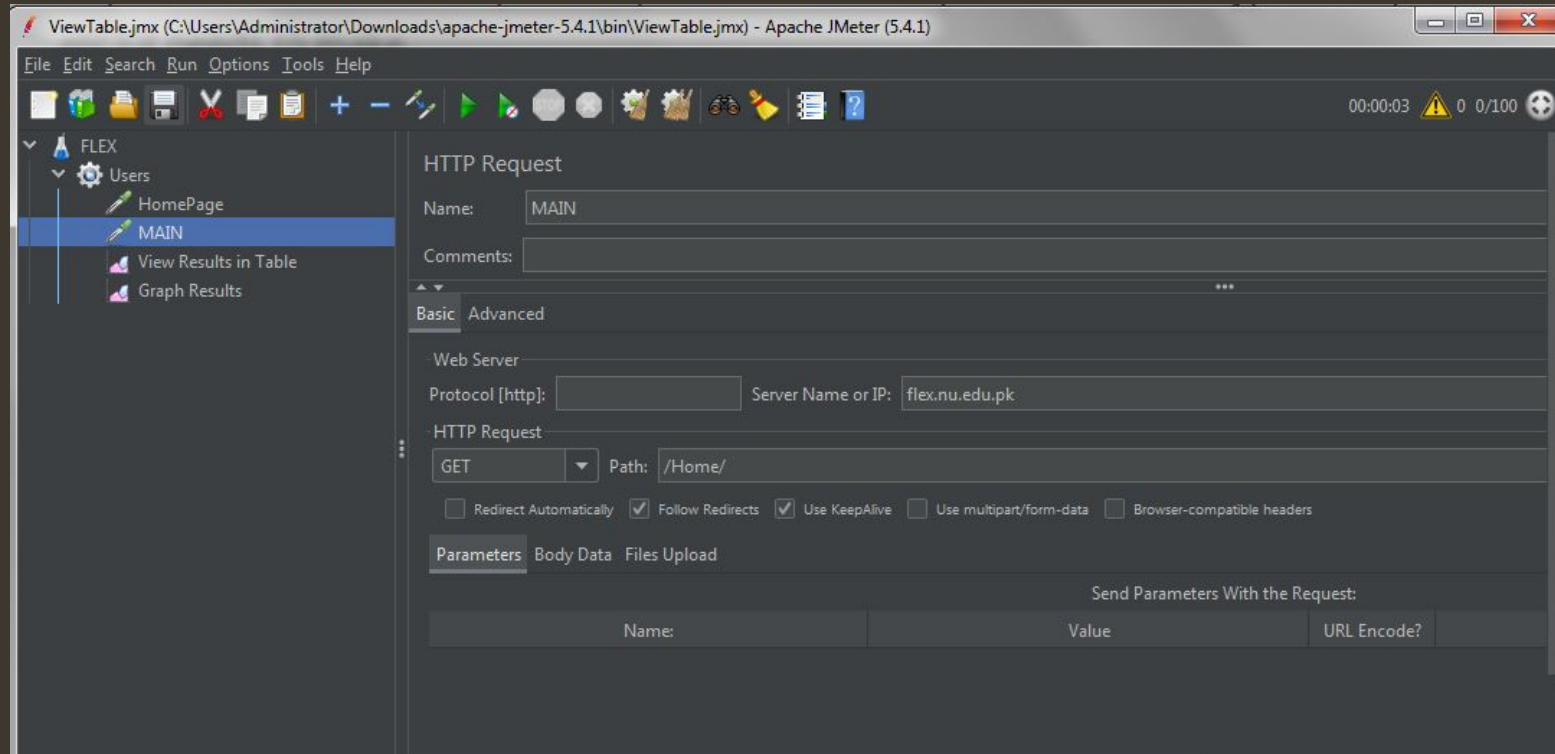
If you are using Window, just run the file **/bin/jmeter.bat** to start JMeter in GUI mode:



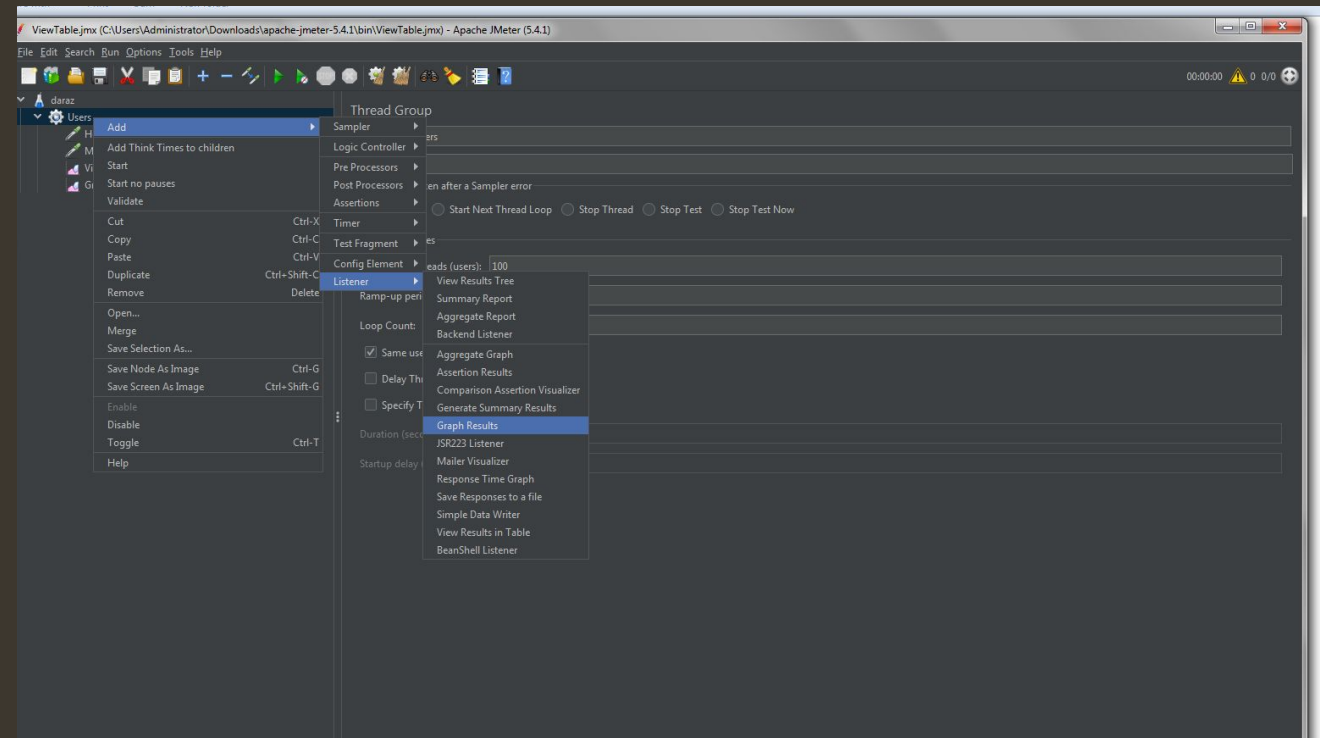
# HomePage



# MAIN







# Result Report

ViewTable.jmx (C:\Users\Administrator\Downloads\apache-jmeter-5.4.1\bin\ViewTable.jmx) - Apache JMeter (5.4.1)

File Edit Search Run Options Tools Help

00:00:03 0 0/100

FLEX

- Users
  - HomePage
  - MAIN
  - View Results in Table**
  - Graph Results

View Results in Table

Name: View Results in Table

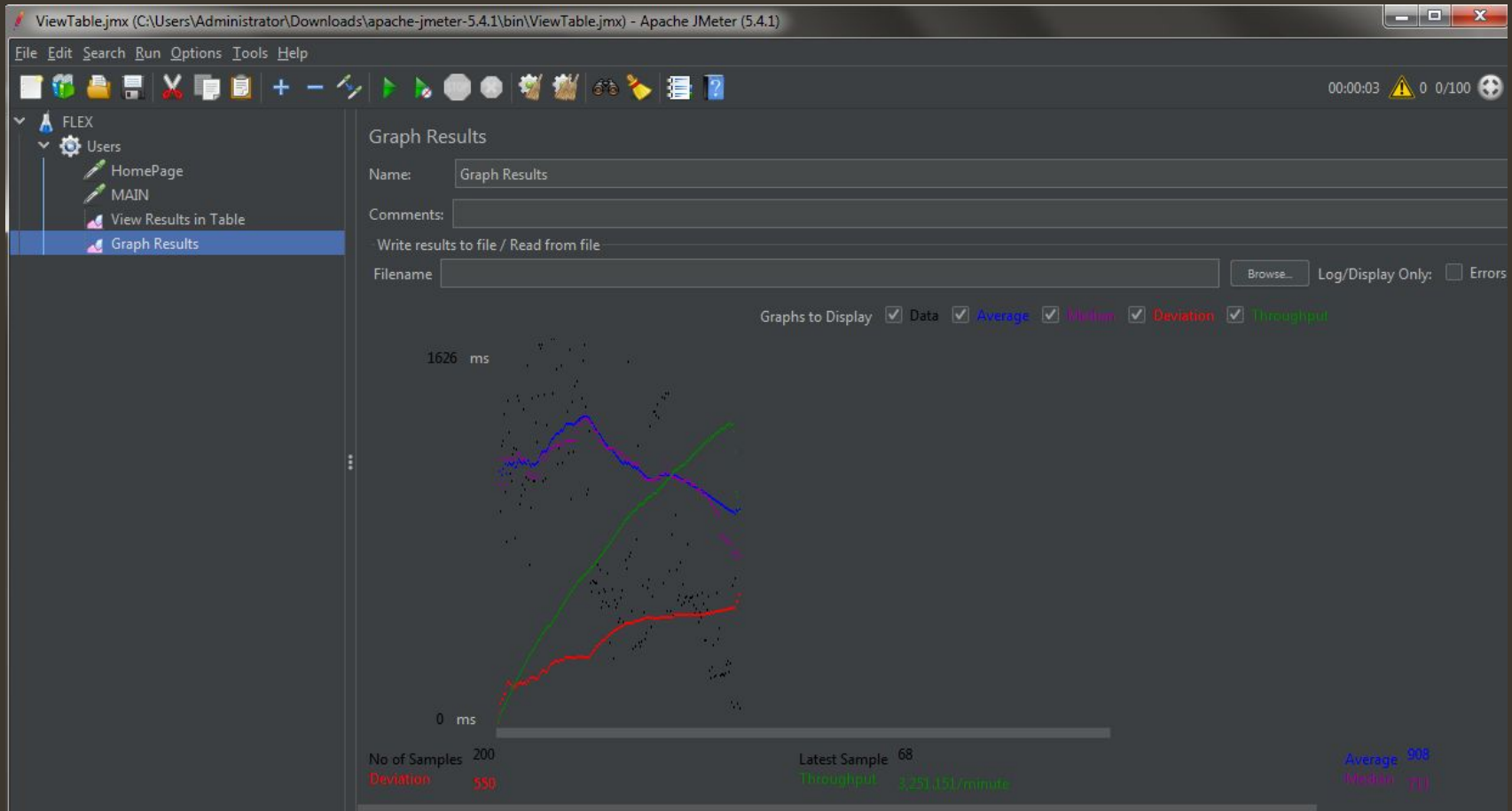
Comments:

Write results to file / Read from file

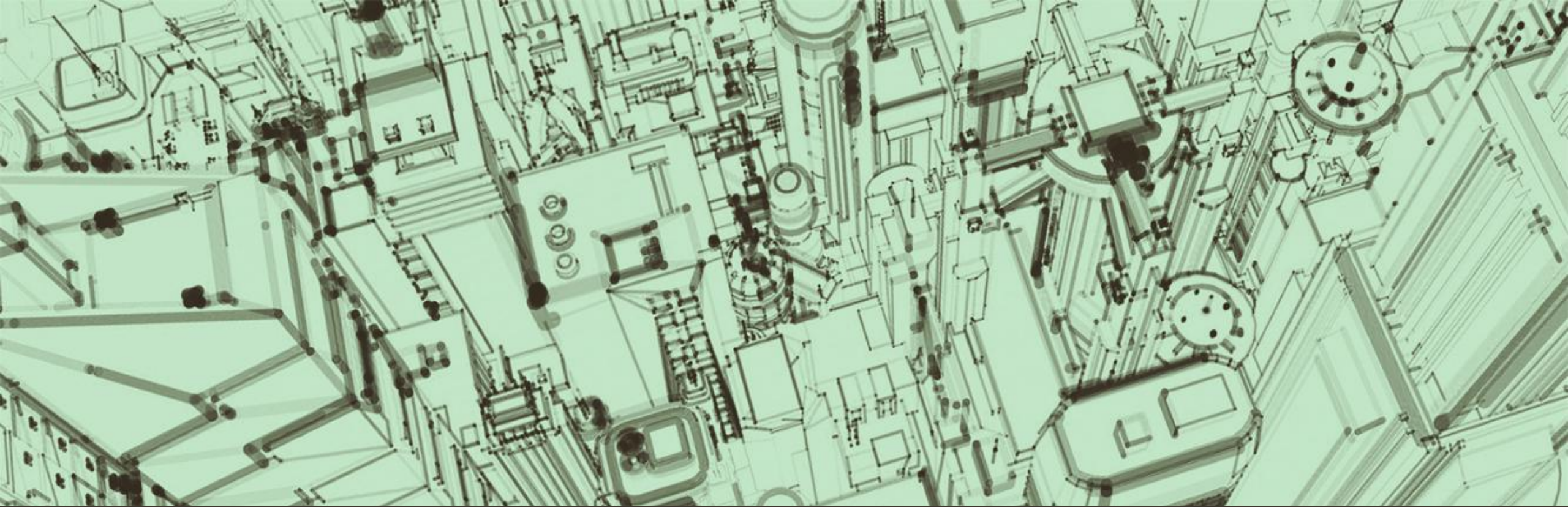
Filename: Browse... Log/Display Only: ☐ Errors ☐ Successes Configure

| Sample # | Start Time   | Thread Name | Label    | Sample Time(... | Status | Bytes | Sent Bytes | Latency | Connect Tim ... |
|----------|--------------|-------------|----------|-----------------|--------|-------|------------|---------|-----------------|
| 1        | 13:47:52.009 | Users 1-30  | HomePage | 1123            | ✓      | 15160 | 238        | 53      | 26              |
| 2        | 13:47:52.121 | Users 1-43  | HomePage | 1014            | ✓      | 15160 | 238        | 56      | 27              |
| 3        | 13:47:52.251 | Users 1-56  | HomePage | 898             | ✓      | 15160 | 238        | 79      | 29              |
| 4        | 13:47:51.977 | Users 1-27  | HomePage | 1175            | ✓      | 15160 | 238        | 56      | 29              |
| 5        | 13:47:51.967 | Users 1-26  | HomePage | 1199            | ✓      | 15160 | 238        | 47      | 21              |
| 6        | 13:47:51.857 | Users 1-15  | HomePage | 1312            | ✓      | 15160 | 238        | 63      | 23              |
| 7        | 13:47:52.191 | Users 1-50  | HomePage | 981             | ✓      | 15160 | 238        | 45      | 23              |
| 8        | 13:47:52.412 | Users 1-72  | HomePage | 770             | ✓      | 15160 | 238        | 47      | 23              |
| 9        | 13:47:51.823 | Users 1-9   | HomePage | 1366            | ✓      | 15160 | 238        | 94      | 52              |
| 10       | 13:47:51.957 | Users 1-25  | HomePage | 1234            | ✓      | 15160 | 238        | 48      | 27              |
| 11       | 13:47:52.141 | Users 1-45  | HomePage | 1057            | ✓      | 15160 | 238        | 57      | 26              |
| 12       | 13:47:52.179 | Users 1-48  | HomePage | 1028            | ✓      | 15160 | 238        | 49      | 24              |
| 13       | 13:47:52.068 | Users 1-36  | HomePage | 1140            | ✓      | 15160 | 238        | 47      | 23              |
| 14       | 13:47:52.048 | Users 1-34  | HomePage | 1167            | ✓      | 15160 | 238        | 44      | 21              |
| 15       | 13:47:52.462 | Users 1-75  | HomePage | 787             | ✓      | 15160 | 238        | 67      | 22              |
| 16       | 13:47:52.091 | Users 1-40  | HomePage | 1162            | ✓      | 15160 | 238        | 53      | 25              |
| 17       | 13:47:51.877 | Users 1-17  | HomePage | 1389            | ✓      | 15160 | 238        | 58      | 27              |
| 18       | 13:47:51.907 | Users 1-20  | HomePage | 1365            | ✓      | 15160 | 238        | 68      | 34              |
| 19       | 13:47:52.356 | Users 1-66  | HomePage | 933             | ✓      | 15160 | 238        | 56      | 26              |

# GRAPH 200th







# Lines of Code and Functional Points



# Software Project Estimating

- Estimation of the size of software is an essential part of Software Project Management. It helps the project manager to further predict the effort and time which will be needed to build the project.
- Various measures are used in project size estimation. Some of these are:
  - LOC
  - Functional Point



# Software Project Estimating

- LOC and FP data are used in two ways during software project estimation:
  - (1) as estimation variables to “size” each element of the software
  - (2) as baseline metrics collected from past projects and used in conjunction with estimation variables to develop cost and effort projections.

LOC and FP estimation are distinct estimation techniques. Yet both have a number of characteristics in common.

Baseline productivity metrics (e.g., LOC/pm or FP/pm<sup>6</sup>)

## LOC- with historical data

A three-point or expected value can then be computed. The *expected value* for the estimation variable (size)  $S$  can be computed as a weighted average of the optimistic ( $s_{\text{opt}}$ ), most likely ( $s_m$ ), and pessimistic ( $s_{\text{pess}}$ ) estimates. For example,

$$S = \frac{s_{\text{opt}} + 4s_m + s_{\text{pess}}}{6} \quad (26.1)$$

- Once the expected value for the estimation variable has been determined, historical LOC or FP productivity data are applied.

# Example of LOC

- The mechanical CAD software will accept two- and three-dimensional geometric data from an engineer. The engineer will interact and control the CAD system through a user interface that will exhibit characteristics of good human/machine interface design. All geometric data and other supporting information will be maintained in a CAD database.
- Design analysis modules will be developed to produce the required output, which will be displayed on a variety of graphics devices. The software will be designed to control and interact with peripheral devices that include a mouse, digitizer, laser printer, and plotter.

# Example Continue

- A range of LOC estimates is developed for each function.
- For example, the range of LOC estimates for the 3D geometric analysis function is
  1. optimistic, 4600 LOC;
  2. most likely, 6900 LOC;
  3. pessimistic, 8600 LOC.

Applying Equation 26.1, the expected value for the 3D geometric analysis function is 6800 LOC.

$$S = \frac{S_{\text{opt}} + 4S_m + S_{\text{pess}}}{6}$$

| Function                                     | Estimated LOC |
|----------------------------------------------|---------------|
| User interface and control facilities (UICF) | 2,300         |
| Two-dimensional geometric analysis (2DGA)    | 5,300         |
| Three-dimensional geometric analysis (3DGA)  | 6,800         |
| Database management (DBM)                    | 3,350         |
| Computer graphics display facilities (CGDF)  | 4,950         |
| Peripheral control function (PCF)            | 2,100         |
| Design analysis modules (DAM)                | 8,400         |
| <i>Estimated lines of code</i>               | 33,200        |

A review of historical data indicates that the organizational average productivity for systems of this type is 620 LOC/pm. And Based on a burdened labor rate of \$8000 per).

1. No of pm =  $33200/620 = 54$  person month.
2. No of LOC/pm =  $33200/54 = 626$  loc/pm.
3. Cost per line =  $\$8000/626 = \$13$
4. Total Cost Project =  $8000*54 = \$ 431,000$



# Function Point

- In this method, the number and type of functions supported by the software are utilized to find FPC(function point count). The steps in function point analysis are:
  1. Count the number of functions of each proposed type. **count**
  2. Compute the Unadjusted Function Points(UFP).
  3. Find Total Degree of Influence(TDI).
  4. Compute Value Adjustment Factor(VAF).
  5. Find the Function Point Count(FPC).

# Count the number of functions of each proposed type

- Find the number of functions belonging to the following types:
  1. External Inputs: Functions related to data entering the system.
  2. External outputs: Functions related to data exiting the system.
  3. External Inquiries: They leads to data retrieval from system but don't change the system.
  4. Internal Files: Logical files maintained within the system. Log files are not included here.
  5. External interface Files: These are logical files for other applications which are used by our system.

| Information Domain Value        | Count                |   | Weighting factor |         |         |   |                      |
|---------------------------------|----------------------|---|------------------|---------|---------|---|----------------------|
|                                 |                      |   | Simple           | Average | Complex |   |                      |
| External Inputs (EIs)           | <input type="text"/> | × | 3                | 4       | 6       | = | <input type="text"/> |
| External Outputs (EOs)          | <input type="text"/> | × | 4                | 5       | 7       | = | <input type="text"/> |
| External Inquiries (EQs)        | <input type="text"/> | × | 3                | 4       | 6       | = | <input type="text"/> |
| Internal Logical Files (ILFs)   | <input type="text"/> | × | 7                | 10      | 15      | = | <input type="text"/> |
| External Interface Files (EIFs) | <input type="text"/> | × | 5                | 7       | 10      | = | <input type="text"/> |
| Count total                     | →                    |   |                  |         |         |   | <input type="text"/> |

To compute function points (FP), the following relationship is used:

$$FP = \text{count total} \times [0.65 + 0.01 \times \sum (F_i)]$$

Value adjustment factors are used to provide an indication of problem complexity.

1. Does the system require reliable backup and recovery?
2. Are specialized data communications required to transfer information to or from the application?
3. Are there distributed processing functions?
4. Is performance critical?
5. Will the system run in an existing, heavily utilized operational environment?
6. Does the system require online data entry?
7. Does the online data entry require the input transaction to be built over multiple screens or operations?
8. Are the ILFs updated online?
9. Are the inputs, outputs, files, or inquiries complex?
10. Is the internal processing complex?
11. Is the code designed to be reusable?
12. Are conversion and installation included in the design?
13. Is the system designed for multiple installations in different organizations?
14. Is the application designed to facilitate change and ease of use by the user?

**0 (not important or applicable) to 5 (absolutely essential)**

The count total shown in Figure 23.3 must be adjusted using Equation (23.1). For the purposes of this example, we assume that  $\Sigma(F_i)$  is 46 (a moderately complex product). Therefore,

$$FP = 50 \times [0.65 + (0.01 \times 46)] = 56$$

Based on the projected FP value derived from the requirements model, the project team can estimate the overall implemented size of the *SafeHome* user interaction

| Information Domain Value        | Count |   | Weighting factor |         |         |   |    |
|---------------------------------|-------|---|------------------|---------|---------|---|----|
|                                 |       |   | Simple           | Average | Complex |   |    |
| External Inputs (EIs)           | 3     | × | 3                | 4       | 6       | = | 9  |
| External Outputs (EOs)          | 2     | × | 4                | 5       | 7       | = | 8  |
| External Inquiries (EQs)        | 2     | × | 3                | 4       | 6       | = | 6  |
| Internal Logical Files (ILFs)   | 1     | × | 7                | 10      | 15      | = | 7  |
| External Interface Files (EIFs) | 4     | × | 5                | 7       | 10      | = | 20 |
| Count total                     |       |   |                  |         |         |   | 50 |



- Assume that past data indicates that one FP translates into 60 lines of code (an object-oriented language is to be used) and that 12 FPs are produced for each person-month of effort. These historical data provide the project manager with important planning information that is based on the requirements model rather than preliminary estimates. Assume further that past projects have found an average of three errors per function point during requirements and design reviews and four errors per function point during unit and integration testing. These data can ultimately help you assess the completeness of your review and testing activities.

| Information domain value           | Opt. | Likely | Pess. | Est.<br>count | Weight | FP<br>count |
|------------------------------------|------|--------|-------|---------------|--------|-------------|
| Number of external inputs          | 20   | 24     | 30    | 24            | 4      | 97          |
| Number of external outputs         | 12   | 15     | 22    | 16            | 5      | 78          |
| Number of external inquiries       | 16   | 22     | 28    | 22            | 5      | 88          |
| Number of internal logical files   | 4    | 4      | 5     | 4             | 10     | 42          |
| Number of external interface files | 2    | 2      | 3     | 2             | 7      | 15          |
| <i>Count total</i>                 |      |        |       |               |        | 320         |

| Factor                                  | Value       |
|-----------------------------------------|-------------|
| Backup and recovery                     | 4           |
| Data communications                     | 2           |
| Distributed processing                  | 0           |
| Performance critical                    | 4           |
| Existing operating environment          | 3           |
| Online data entry                       | 4           |
| Input transaction over multiple screens | 5           |
| Master files updated online             | 3           |
| Information domain values complex       | 5           |
| Internal processing complex             | 5           |
| Code designed for reuse                 | 4           |
| Conversion/installation in design       | 3           |
| Multiple installations                  | 5           |
| Application designed for change         | 5           |
| <b>Value adjustment factor</b>          | <b>1.17</b> |

$$FP = \text{Count-total} * [0.65 + 0.01 * \sum(f_i)]$$

$$320 * [0.65 + 0.01 * 52]$$

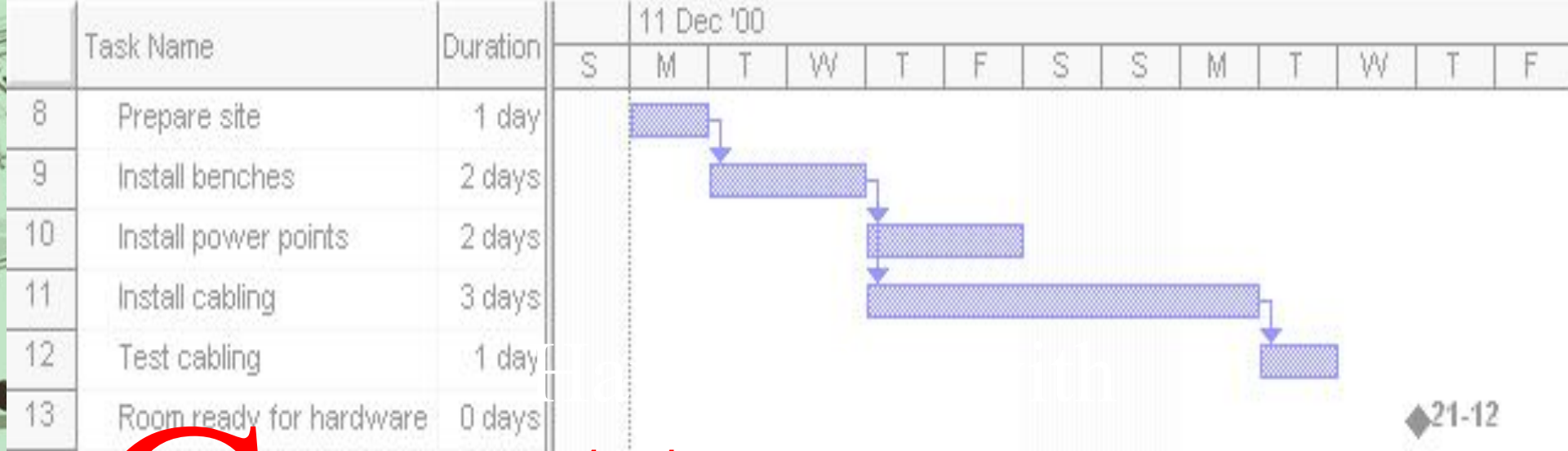
$$FP = 320 * [1.17]$$

*\*Given*

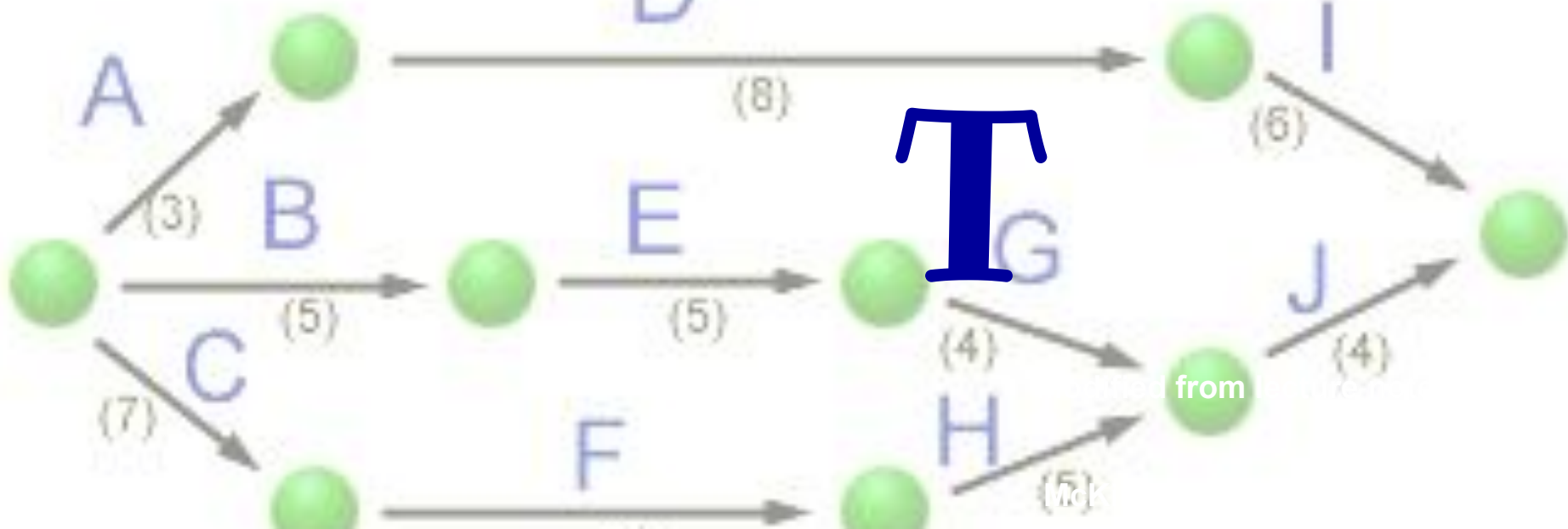
$$FP = 375$$

The organizational average productivity for systems of this type is 6.5 FP/pm.

- **Cost Per FP = 8000/6.5**  
Based on a burdened labor rate of \$8000 per month, the cost per FP is approximately \$1230.
- **Project Cost = 1230\*375 = 461000\$**  
Based on the FP estimate and the historical productivity data, the total estimated project cost is \$461,000
- **Effort= FP/ 6.5 FP/pm= 375/6.5=58**  
Estimated effort is 58 person-months.



# Gantt & PERT

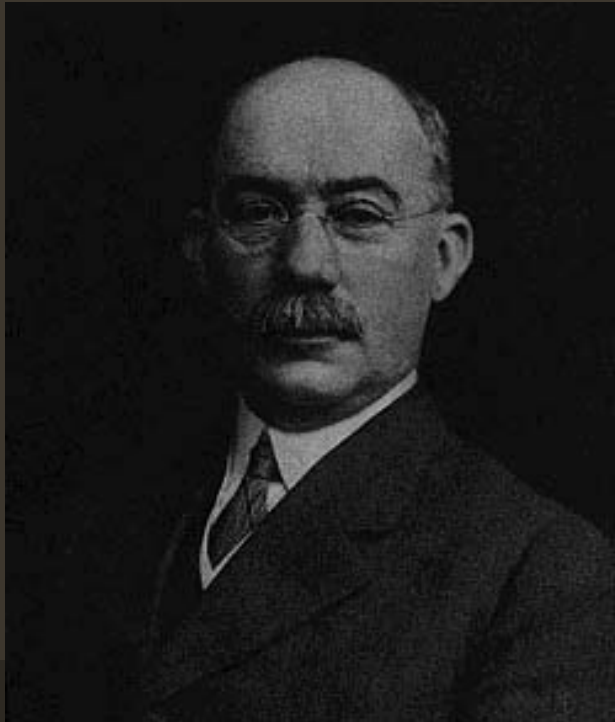


# What are they?

Gantt and PERT charts are both “CPM” (Critical Path Method) tools to:

- manage the tasks involved in big and complex projects
- let project managers organise time, people, equipment and money
- ensure the right people and equipment are in the right place and the right time
- allow managers to monitor the progress of a project

# Gantt Charts



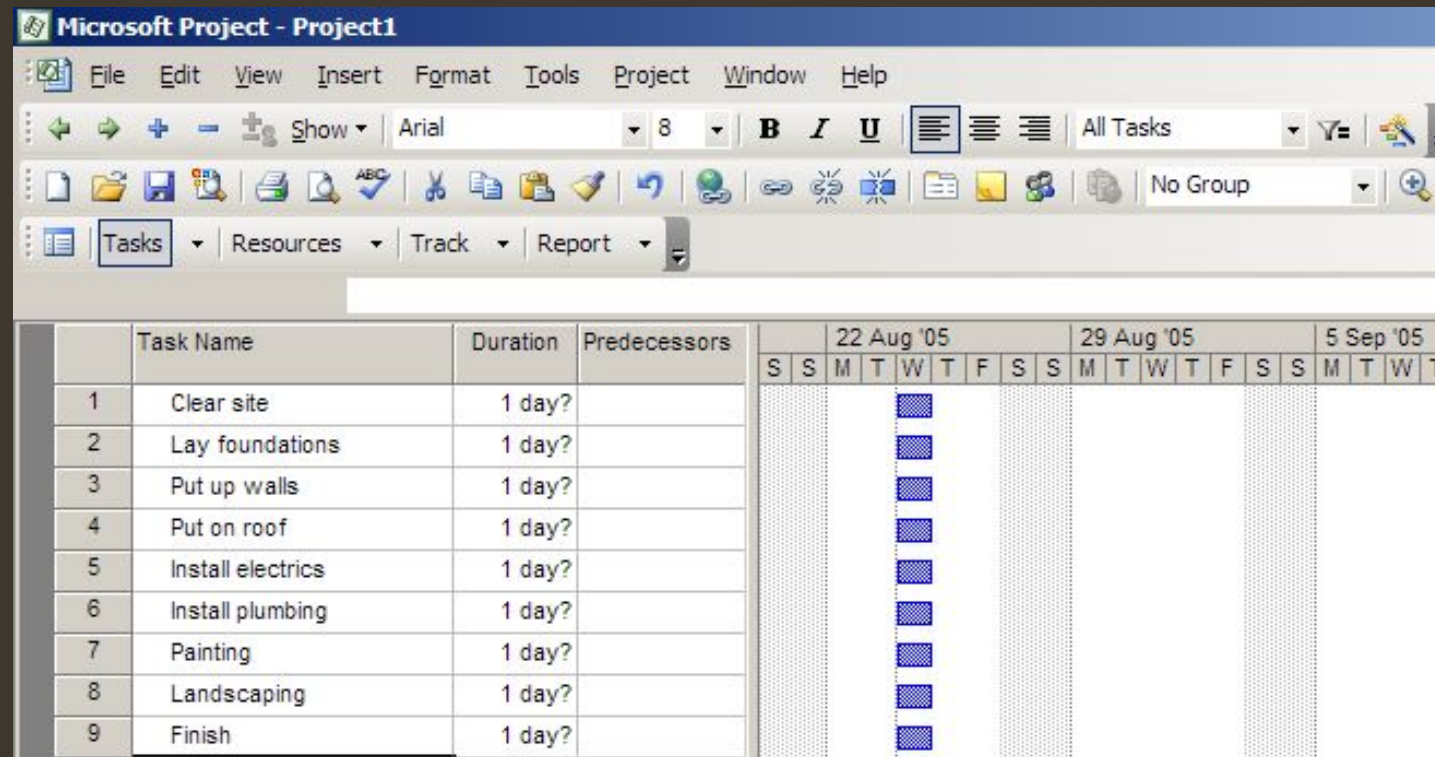


# Gantt Basics

- Basically, a timeline with tasks that can be connected to each other
- Note the spelling!
- It is not all-capitals!
- Can be created with simple tools like Excel, but specialised tools like Microsoft Project make life easier

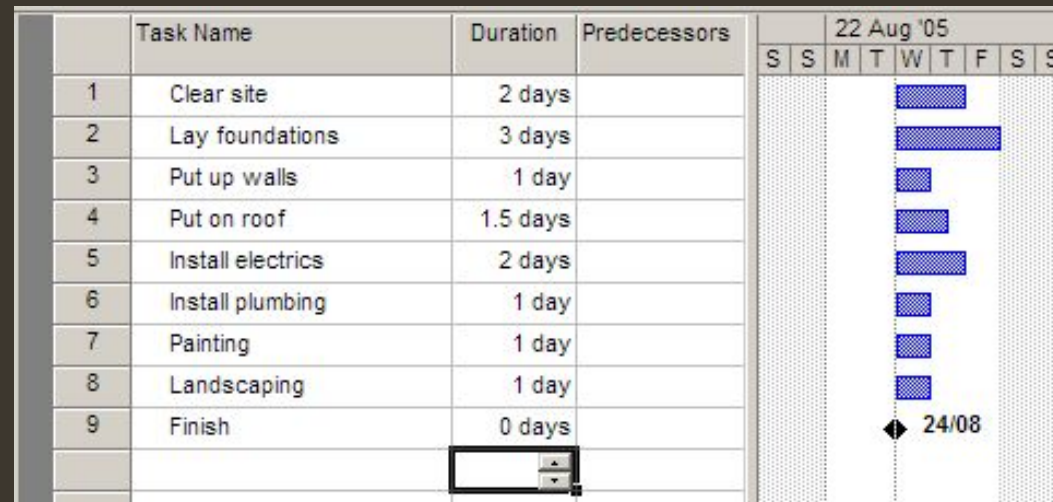
# Making a Gantt chart

- Step 1 – list the tasks in the project



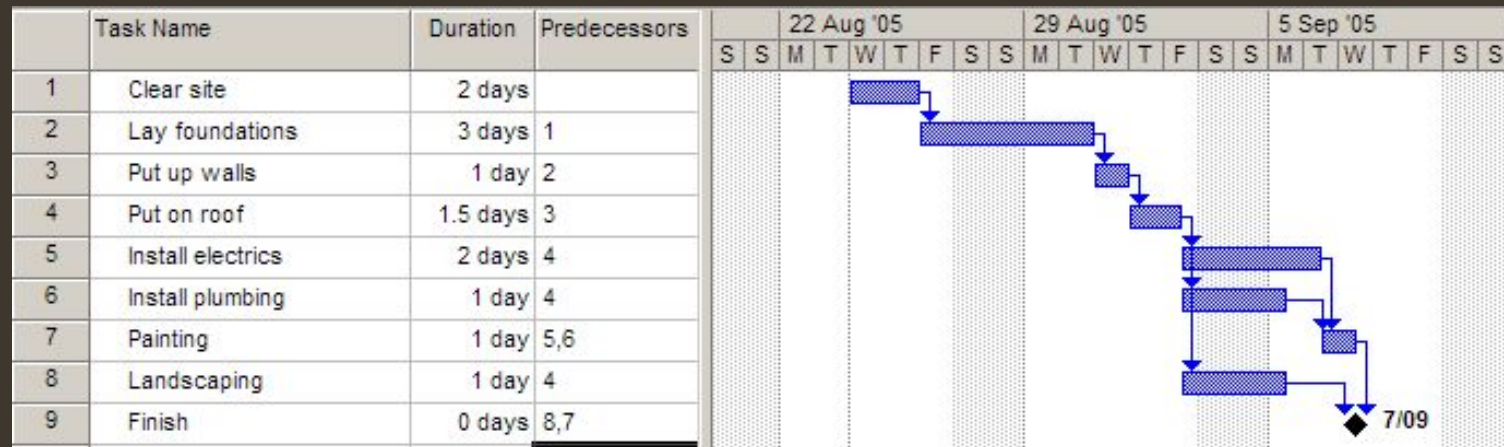
# Making a Gantt chart

- Step 2 – add task durations

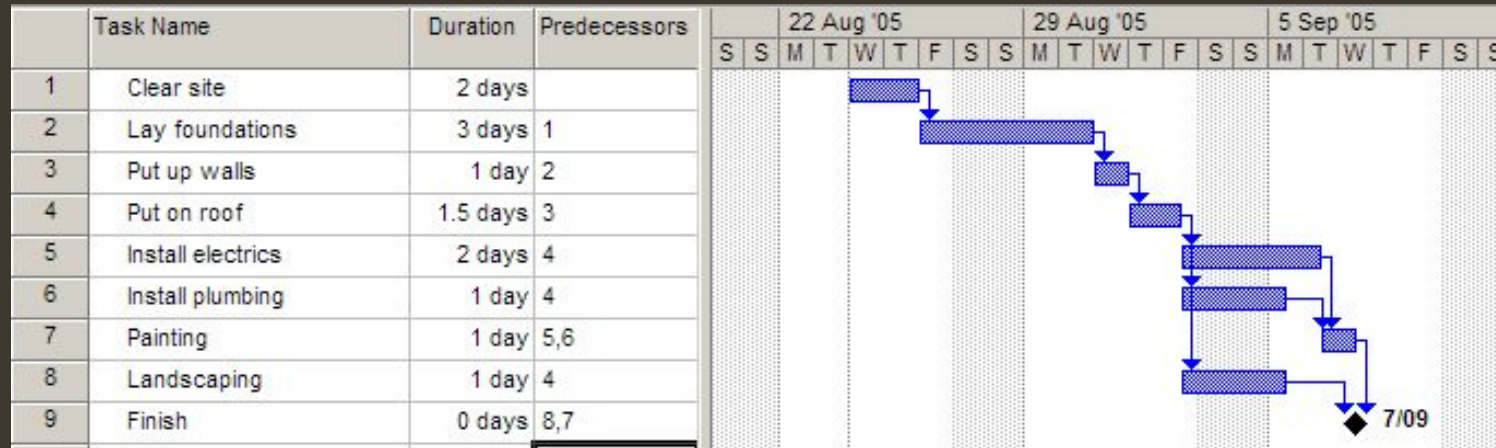


# Making a Gantt chart

Step 3 – add *dependencies* (which tasks cannot start before another task finishes)



# Notes

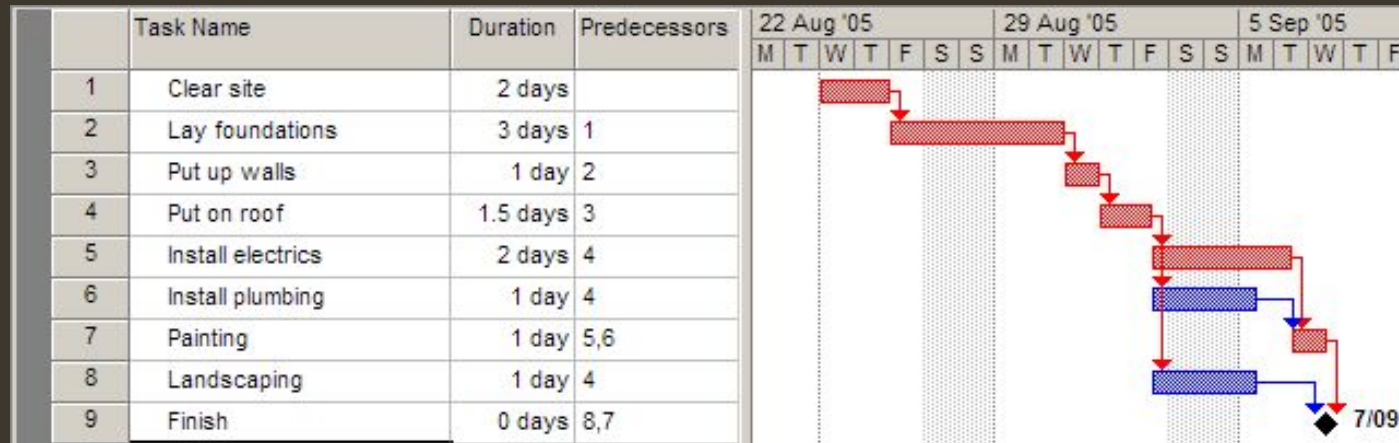


- The arrows indicate **dependencies**.
- Task 1 is a **predecessor** of task 2 – i.e. task 2 cannot start before task 1 ends.
- Task 3 is **dependent** on task 2. Task 7 is dependent on two other tasks
- Electrics, plumbing and landscaping are **concurrent** tasks and can happen at the same time, so they overlap on the chart. All 3 can start after task 4 ends.
- Painting must wait for both electrics and plumbing to be finished.
- Task 9 has zero duration, and is a **milestone**



# Making a Gantt chart

- Step 4 – find the **critical path**

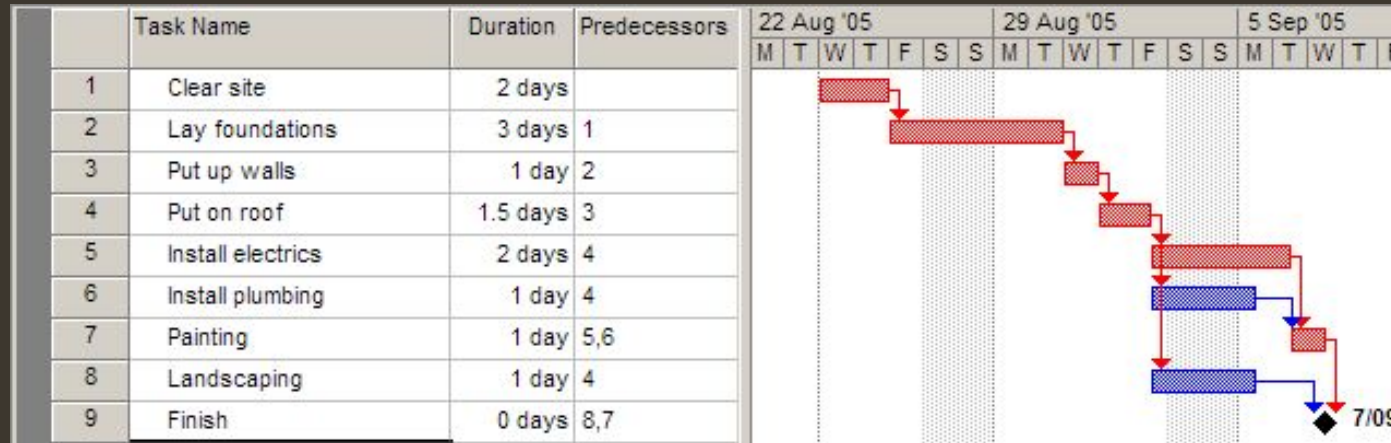


The critical path is the sequence of tasks from beginning to end that takes the **longest time** to complete.

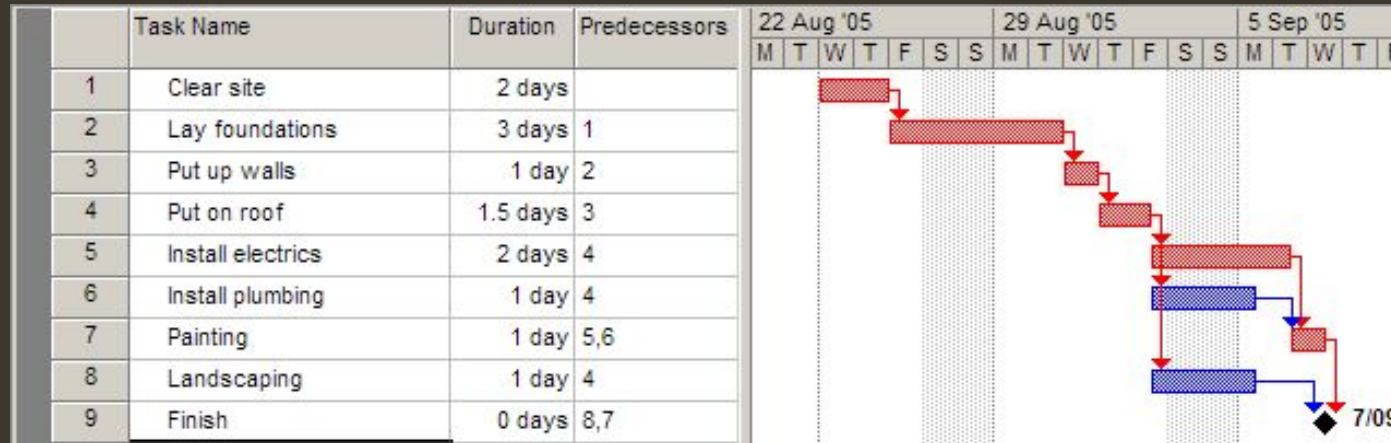
It is also the **shortest possible time** that the project can be finished in.

Any task on the critical path is called a **critical task**.

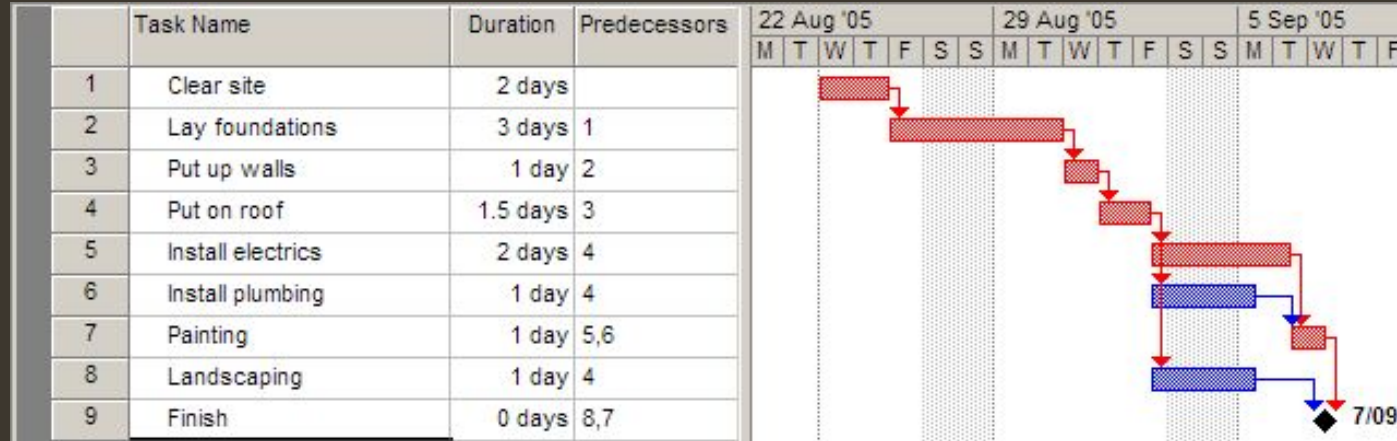
No critical task can have its duration changed without affecting the end date of the project.



- MS Project can work out the critical path for you!
- The length of the critical path is the sum of the lengths of all critical tasks (the red tasks 1,2,3,4,5,7) which is  $2+3+1+1.5+2+1 = 10.5$  days.
- In other words, the minimum amount of time required to get all tasks completed is 10.5 days
- The other tasks (6,8) can each run over-time before affecting the end date of the project



- The amount of time a task can be extended before it affects other tasks is called *slack* (or *float*).
- $Float = Late\ Start - Early\ Start\ OR\ Late\ Finish - Early\ Finish$
- Task 6 can take an extra day and a half before it affects the project's end date, so each has **1.5 day's slack**.



Critical tasks, by definition, can have NO slack.

Thus if you are ever asked, “*Can the duration of a critical task be changed without affecting the end date of the project?*”, the answer is always **NO!**